Self-Adjointness of the Fundamental Fractal Operator $\hat{F}$

Closure of the R82 gap from Doc. 322

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August 2026

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Status markers: **[B]** algebraically proved; **[K]** derivable from ξ or numerically checked; **[S]** sketched or postulated, not proved (open).

Abstract

Doc. 322 defines the fundamental fractal operator

$$\hat{F} = \bigoplus_{n=1}^{N} S_n \circ L_n, \qquad S_n = r_n U_n, \quad r_n \in (0,1), \quad U_n \text{ unitary}, \tag{.1}$$

with logical projectors $L_n^2 = L_n$, $L_n^\dagger = L_n$, $L_n L_m = L_{\min(n,m)}$, and books self-adjointness as an open sketch **[S]** (R82).

The upper bound N in (.1) is *finite*: $L_0 = \xi \cdot \ell_P$ (Doc. 180 **[K]**) is the minimal length scale of FFGFT — below L_0 the concept of "distance" loses physical meaning. The $\bigoplus_{n=1}^{\infty}$ notation in Doc. 322 is a functional-analytic convention, not a physical infinity.

Main result [B] on P_0, [S] on all of $\mathscr{H}$

The R82 gap is **closed on the $\mathbb{Z}_3$-invariant sector** P_0; on the whole space it remains open **[S]**. Because L_0 makes the number of rungs finite, $\hat{F}$ is bounded:

$$\|\hat{F}\| \;\le\; \sum_{n=1}^{N} r_n \;<\; \infty.$$

A bounded symmetric operator is automatically self-adjoint; the deficiency indices are $(0,0)$; there is exactly one self-adjoint realisation. The $\mathbb{Z}_3$ pairing $(k,-k)$ makes $\hat{F}$ normal; on P_0, $\hat{F} = \hat{F}^\dagger$. On charged pairs $\hat{F}^\dagger$ is the $\mathbb{Z}_3$-mirrored $\hat{F}$; the self-adjoint operator there is $\mathrm{Re}\,\hat{F} = (\hat{F} + \hat{F}^\dagger)/2$ — a postulate **[S]**. $\xi = \lambda_{\min}(\hat{F}_{D_4})$ is well-defined provided the D_4 sector lies in the $\mathbb{Z}_3$-invariant sector **[B]**.

Verification script: `327_selbstadjungiertheit_F_verify.py` — 26 assertions, all passed.

.1 Starting point

R82 (Doc. 190) records: *"Open: complete self-adjointness proof for $\hat{F}$* ***[S].*** *"* Without this proof, the central claim

$$\xi = \lambda_{\min}(\hat{F}_{D_4}) = \frac{4}{30000}$$

is formally unsecured: "smallest eigenvalue" requires a real spectrum, and a real spectrum requires self-adjointness.

The definition in Doc. 322 contains a genuine tension: $S_n L_n = r_n U_n L_n$ is in general *not* symmetric for arbitrary unitary U_n — $(U_n L_n)^\dagger = L_n U_n^\dagger \neq U_n L_n$. The self-adjointness must come from the *structure*. On the $\mathbb{Z}_3$-invariant sector P_0, this is what the $\mathbb{Z}_3$ pairing delivers.

.2 Three observations from the corpus

The following points introduce no new axioms — they read from already-established corpus results what is needed for the proof.

(F1) Finite number of rungs. $L_0 = \xi \cdot \ell_P$ is the minimal length scale (Doc. 180 **[K]**). The scale ladder ends at L_0; N is finite; $\sum_{n=1}^{N} r_n < \infty$ follows trivially. No convergence condition needs to be imposed — there are no infinities in FFGFT.

(F2) $\mathbb{Z}_3$ phase pairing. U_n acts sector-wise as a phase rotation $e^{i\theta_{n,k}}$ (Doc. 322: "phase rotations on the scale sectors"), and the phases respect the orbifold symmetry: $\theta_{n,-k} = -\theta_{n,k}$ for $\mathbb{Z}_3$-charged pairs $(k, -k)$; on null sectors ($q = 0$): $\theta \in \{0, \pi\}$. The same pairing carries the Hawking sector information in Doc. 325.

(F3) Finite fractal recursion. The fractal measure arises from a finite 100-step recursion (Doc. 322), hence $0 < c \leq w \leq C$ for the weight w — bounded and strictly positive.

.3 Proof chain

Lemma 1 — Filtration realises the Doc.-322 axioms [B]

Under (F1), the L_n satisfy exactly $L_n^2 = L_n$, $L_n^\dagger = L_n$, $L_n L_m = L_{\min(n,m)}$: the "fractal inclusion structure" of Doc. 322 is a scale filtration. *(Script [1]: exact to machine precision.)*

Theorem 2 — Diagonal form and boundedness [B]

Under (F1)+(F2), $\hat{F}$ is block-diagonal:

$$\hat{F}\big|_{\mathscr{H}_k} = \varphi_k \cdot \mathbb{1}, \qquad \varphi_k = \sum_{n \geq k}^{N} r_n \, e^{i\theta_{n,k}}.$$

By the finite rung count (F1): $\|\hat{F}\| \le \sum_{n=1}^{N} r_n < \infty$. $\hat{F}$ is **bounded**, $\mathscr{D}(\hat{F}) = \mathscr{H}$. *(Script [2]+[3].)*

Theorem 3 — $\mathbb{Z}_3$ pairing: normality [B], symmetry on P_0 [B]

Under (F2), $\hat{F}$ is normal (block-diagonal with scalars φ_k); $\varphi_{-k} = \bar{\varphi}_k$ holds when the sectors k and $-k$ carry the same scale rungs n. On the null sectors ($\theta \in \{0, \pi\}$), $\hat{F}$ is self-adjoint. On charged pairs $\hat{F}^\dagger$ is the $\mathbb{Z}_3$-mirrored $\hat{F}$, $\|\hat{F} - \hat{F}^\dagger\| = 2|\mathrm{Im}\,\varphi_k|$; the self-adjoint operator there is

$$\mathrm{Re}\,\hat{F} = \tfrac{1}{2}(\hat{F} + \hat{F}^\dagger), \qquad \varphi_k^{\mathrm{cov}} = \sum_{n \ge k}^{N} r_n \cos\theta_{n,k} \in \mathbb{R}$$

(cosine form) — a postulate **[S]**. *(Script [4]: normality, $\|\hat{F} - \hat{F}^\dagger\| = 0.019$ on charged pairs, hermiticity on P_0.)*

Corollary 4 — Self-adjointness, deficiency indices $(0, 0)$ [B] on P_0

Bounded (Thm. 2) + symmetric (Thm. 3) ⇒ self-adjoint: for $\hat{F}$ on $P_0\mathscr{H}$, for $\mathrm{Re}\,\hat{F}$ on all of $\mathscr{H}$ (there a postulate **[S]**). Deficiency indices: $\ker(A \mp i) = \{0\}$ for $A = \hat{F}|_{P_0\mathscr{H}}$ or $A = \mathrm{Re}\,\hat{F}$, since $\|(A \mp i)^{-1}\| \le 1$ for bounded self-adjoint A. **Exactly one** self-adjoint realisation — no extension choice. *(Script [5]: smallest singular value exactly 1, for $\mathrm{Re}\,\hat{F}$.)*

Theorem 5 — Tensor factor $\hat{\Delta}_{T^4}$ with fractal measure [B]

1. Δ on compact T^4 (no boundary): essentially self-adjoint, Fourier eigenbasis with eigenvalues $|n|^2 \ge 0$. (Standard.)
2. Fractal measure (F3): $0 < c \le w \le C$, so $\psi \mapsto \sqrt{w}\psi$ is a unitary isomorphism $L^2(d\mu_f) \to L^2(d\mu)$. The fractal measure *preserves* self-adjointness. *(Script [6].)*

Theorem 6 — $\mathbb{Z}_3$ restriction and χ-twists [B]

1. $[\hat{F}, P_0] = 0$ ($\mathbb{Z}_3$ covariance from (F2)) ⇒ restriction to the orbifold sector is self-adjoint. *(Script [7].)*
2. Fixed-point boundary conditions $\chi^3 = 1$ (Docs. 322/320): unitarily implemented twists; spectrum $(n + \alpha)^2 \ge 0$ real in all three twist classes — each carries a self-adjoint realisation. *(Script [8].)*

Corollary 7 — ξ as well-defined minimum [B]

$\hat{F}_{D_4}$ (restriction to the D_4 sector, Doc. 314) is self-adjoint, provided the D_4 sector lies in the $\mathbb{Z}_3$-invariant sector $P_0\mathscr{H}$ (otherwise the statement holds for $\mathrm{Re}\,\hat{F}$, postulate **[S]**). By the min-max principle:

$$\xi = \lambda_{\min}(\hat{F}_{D_4}) = \min_{\|\psi\|=1,\, \psi \in \mathscr{H}_{D_4}} \langle \psi, \hat{F}\psi \rangle$$

well-defined and truncation-stable: the lowest mode lies in every truncation space that contains it. *(Script [9]: ξ is inserted in the model; what is checked is existence and truncation stability of the minimum.)*

.4 Status table

Statement	Status
$L_n L_m = L_{\min}$ realised by filtration, exactly	**[B]**
$\hat{F}$ bounded by finite rung count (L_0)	**[B]**
$\hat{F}$ normal; symmetry on P_0 as $\mathbb{Z}_3$ consequence	**[B]**
Symmetry on the whole space (Re $\hat{F}$ as postulate)	**[S]**
Self-adjointness, deficiency indices $(0,0)$, uniqueness on P_0	**[B]**
Fractal measure (finite recursion) preserves self-adjointness	**[B]**
$\mathbb{Z}_3$ restriction and all three χ-twist classes s.a.	**[B]**
$\xi = \lambda_{\min}(\hat{F}_{D_4})$ well-defined, truncation-stable (D_4 in P_0)	**[B]**
Remaining cases	whole space open **[S]**

The R82 gap is closed on the $\mathbb{Z}_3$-invariant sector; on the whole space it remains open **[S]**. There are no infinities in FFGFT; $\sum r_n < \infty$ was never an extra requirement — it follows directly from $L_0 = \xi \cdot \ell_{\mathrm{P}}$.

Relation to the corpus

Building block	Doc.
Minimal length scale $L_0 = \xi \cdot \ell_{\mathrm{P}}$	180
Definition of $\hat{F}$, $\mathscr{D}(\hat{F})$, R82 gap	322
D_4 lattice in Hilbert space	314
Fixed-point boundary conditions, neutrino localisation	320, 322
$\mathbb{Z}_3$ triality, sector structure	321
Sector pairing $(k, -k)$ — same as in Hawking mechanism	325
Fractal measure, 100-step recursion, $D_f = 3 - \xi$	322

References

1. J. Pascher, Docs. 180, 314, 320, 321, 322, 325, FFGFT corpus, 2026.
2. M. Reed, B. Simon, *Methods of Modern Mathematical Physics I/II*, Academic Press.
3. J. von Neumann, *Allgemeine Eigenwerttheorie Hermitescher Funktionaloperatoren*, Math. Ann. 102, 1929.

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